

H0	H1	H2	H3	H4	H5
Starting bias	Meaning across depth	Readout validation	Context and anchor	Decoder-anchor revision	Meaning revision
A model may be biased towards one sense of a homonym because of its training data. If a model consistently chooses one sense, choosing that sense is not meaningful evidence in the later hypotheses.	Language processing may be hierarchical in the model. If early layers mainly preserve word-form information, the sense groups should overlap for both homonyms; if contextual meaning emerges with depth, they should increasingly separate. If such a hierarchy emerges, there should be a single layer or a region of layers in which semantic/contextual processing dominates.	If candidate semantic layer(s) can be identified, their role must be validated by testing whether they also classify unseen sentences and unseen homonyms.	If such semantic layer(s) exist, token representations in semantic space should show the model moving from an ambiguous homonym state towards either sense once resolving context is added. This should differ between encoders and decoders because they have different access to context.	Because causal decoders cannot access subsequent tokens, are the estimated starting biases affected by the homonym’s position in the sentence? H1 is repeated at the end-of-sentence token rather than the homonym position to test whether the identified semantic layer(s) change.	H3 tests ambiguity resolution; H5 tests whether a model can revise an interpretation encouraged by earlier context when later context requires the alternative sense.
CORRECTS results from H1-H5	IDENTIFY SEMANTIC LAYER if it exists	TEST H1 result	TEST H1 result	REVISE H1 result	TEST H1 result
Tests increase in specificity: - If H1 identifies a set of layers, H2-H5 may help narrow them to fewer candidates - If one layer dominates across all tests, it may contain a robust representation of contextual semantic processing					
Quick View of Results					
Priors exist for all models and homonyms, but they differ between isolated bare words and words presented in context.	A clear depth-wise hierarchy emerges. H1 is supported: in the geometric representation of token-level activations, the two senses begin in similar positions and diverge into sense clusters as context is added. However, semantic/contextual information dominates token-level activations quite quickly and, in some cases, plateaus. There is no single semantic layer, although some models may have broader semantic regions. For most models, these regions appear between mid-depth and three-quarter depth.	Candidate layers, including the selected winners, generalise. However, different metrics produce different winners and, in some cases, different candidate regions.	For decoders, the geometry shows a clear transition from an ambiguous shared location to distinct sense regions once the resolver is added. For encoders, left and right resolution make little difference. The geometric evolution suggests that sentence-level ambiguity moves the token-level representations closer together regardless of resolver order. This is consistent with encoders having access to all tokens.	For decoders, H4 produces the same candidate regions and winning layers as H1. The short homonym-in-context sentences therefore appear sufficiently balanced not to skew candidate generation.	H5 consistently shows geometric movement from one sense towards the other, although revision does not always cross the decision boundary. The primed sense persists with different strength depending on the homonym and model. This suggests a substantial cost of revision rather than a clean overwrite.
Methods					
- Activation collection: bare word vs. carrier - Centroid probing: two-sense geometry - Sense-margin scoring: prior direction - Carrier consistency: baseline stability	- Layer-wise activations - Centroid separation: sense distance - PCA trajectories: geometric progression - Candidate-layer selection: semantic region	- Nested leave-one-out: sentence generalization - Held-out homonym test: homonym generalisation - Adequacy: item-level accuracy - GDV: global class separation - layer comparison	- Resolver reordering: left- vs. right-context - PCA trajectories: ambiguity to resolution - Correct-sense margin: movement direction - Paired bootstrap: order effect	Position-specific centroids: endpoint vs. homonym position sense-geometry	- Garden-path priming - Fixed-sentinel probing: constant readout for controls - Trajectories: prime to resolver - Matched controls - Boundary crossing